

Defending Intellectus' carbon story

A measured retrofit for an n8n-based social-impact
diagnostic workflow

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Audience: CTO / Operations Lead

A useful report should not pay twice for the same evidence

Intellectus turns public evidence into a consultant-reviewed diagnostic. Its value comes from synthesis and judgment - not from repeating search, extraction, context, or model work.

COST

Fewer paid calls and tokens

LATENCY

Faster reports for consultants

RISK

Less retained and transferred data

One unit of value is one review-ready diagnostic report

R

One completed, evidence-grounded diagnostic report for one social-impact organization, ready for consultant review.

SCI thinking: (Energy × grid intensity + embodied emissions) per R

This makes every proposed change testable per report - not per vague 'AI use.'

The audit follows the verified path, not the future vision

Current repository evidence: live parent workflow + successful child workflow

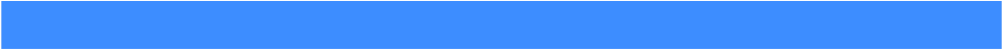
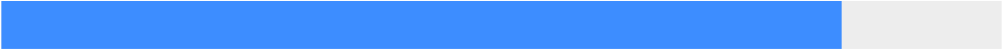


Included in pilot boundary

Search, extraction, orchestration, inference, payload transfer, temporary storage, and execution region.

Likely waste concentrates before the final report exists

Hotspot ranking is directional until telemetry is added

- | | | | |
|----------|---|--|---|
| 1 | Repeated web search and extraction |  | same organizations and public pages may recur |
| 2 | Multi-step orchestration / model calls |  | validation or formatting may not need a model |
| 3 | Long context and duplicated evidence |  | tokens and payload can grow with every branch |
| 4 | Storage, retention, and idle capacity |  | current production settings are unknown |

Repeated work is the cheapest place to start because it reduces compute without changing the diagnostic promise.

The strongest conclusion: measure first

WHAT WE KNOW

- Primary stack is n8n
- Workflow is event-triggered
- Public research and local RAG are used
- Human review is mandatory
- A live 16.38 s parent run is retained

WHAT WE DO NOT KNOW YET

- **Model/provider and model size in production**
- **Calls and tokens per R**
- **Hosting region and grid intensity**
- **Energy or hardware utilization**
- **Cache behavior and retention windows**

Skeptic: ‘Why should we believe the hotspots?’ Answer: treat them as testable hypotheses and instrument the verified boundary.

Four pillars turn a vague green claim into operating choices

CARBON

Where and when electricity is consumed

Region + grid intensity per R

ENERGY

How much compute, transfer, and storage each report requires

Calls, tokens, MB, latency

HARDWARE

How efficiently provisioned resources are used

Idle capacity + model/runtime fit

MEASUREMENT

How changes are compared without greenwashing

Boundary, R, telemetry, quality

Cache evidence, not conclusions

GSF pattern: Cache static data

PROBLEM

The same public page can be searched, downloaded, and extracted again across runs.

CHANGE

Key extracted evidence by normalized URL + content hash + freshness window. Reuse unchanged pages, while keeping timestamps and a manual refresh path.

BUSINESS RESULT

Fewer external calls, lower latency, and a clearer evidence trail.

Pillars: Energy · Hardware · Measurement

Route before invoking a large model

GSF: Optimize agent orchestration · Right-size AI models

PROBLEM

Validation, formatting, or simple routing may trigger expensive inference without adding diagnostic value.

CHANGE

Use deterministic code first. Route extraction/classification to a smaller model, evaluate a representative sample, and fall back only when confidence or quality fails.

BUSINESS RESULT

Lower cost and energy per R without blind quality loss.

Pillars: Energy · Hardware · Measurement

Move and retain only the evidence the report needs

GSF pattern: Reduce transmitted data

PROBLEM

Raw pages, repeated snippets, long prompts, and verbose intermediate outputs increase transfer and storage.

CHANGE

Pass cited snippets instead of whole pages, reuse structured claims, cap context, compress payloads where useful, and define retention by artifact type.

BUSINESS RESULT

Fewer tokens and MB per report, plus lower privacy and debugging exposure.

Pillars: Energy · Hardware · Measurement

Use on-demand execution and test greener placement

GSF: On-demand execution · Carbon-aware scheduling

PROBLEM

Unknown hosting and idle-capacity choices can dominate a low-volume consultant workflow.

CHANGE

Preserve webhook/event-driven execution, avoid always-on workers where feasible, and compare eligible regions or time windows for non-urgent research.

BUSINESS RESULT

Less idle capacity and a defensible region decision balancing carbon, latency, and compliance.

Pillars: Carbon · Energy · Hardware

Observe first, remove waste second, optimize last

WEEK 1

WEEK 2

NEXT SPRINT



Instrument

Log calls, tokens, cache eligibility, payload, latency, cost, region, and quality per R.

Remove repeated work

Cache evidence, deduplicate URLs, add conditional routing, and cap unnecessary context.

Right-size and place

Run quality-backed model comparisons and assess on-demand / carbon-aware infrastructure.

For 1-2 weeks, measure efficiency and quality together

METRIC	PER R	SUCCESS SIGNAL
Research + model calls	count	≥25% fewer repeat fetches; no deterministic step routed to a model
Input/output tokens	tokens	Lower median with quality held inside agreed tolerance
Cache reuse	% eligible	≥30% hit rate on recurring organizations or sources
Payload + retention	MB + days	Lower transfer; every artifact has an explicit retention rule
Latency + cost	seconds + EUR	No material regression; lower median cost
Quality + provenance	pass/error rate	No citation-coverage or factuality decline
Region + grid intensity	location + factor	Every measured run has a disclosed region and source

A successful pilot removes work without weakening quality

BEFORE

Fetch → extract → send full context → model → repeat

Unknown calls, tokens, region, and retention per report

AFTER

Reuse evidence → route conditionally → send cited snippets → review

Pilot targets: ≥25% fewer repeat fetches and ≥30% cache hits, with quality held constant

Hypothesis - not observed performance

Proceed with a measured retrofit pilot

WHAT WE CAN DEFEND

- **R and the first measurement boundary**
- **Likely hotspots tied to real architecture**
- **Four practical GSF-aligned changes**
- **A quality-protected measurement plan**

WHAT WE CANNOT CLAIM YET

- A verified SCI score
- Carbon neutrality
- A greener region without data-residency review
- Emissions savings from token or cost proxies alone

Decision requested: approve 1-2 weeks of telemetry and one controlled before/after retrofit.